

KEUN-YOUNG YOOK

Ph.D. Student, Yonsei University | Visiting Researcher, Harvard Medical School

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EDUCATION

Yonsei University	2024 - Present
Integrated M.S./Ph.D. Program, Electrical and Electronic Engineering	
Yonsei University	2021 - 2024
B.S., School of Integrated Technology	

ACADEMIC APPOINTMENTS

Harvard Medical School	Aug 2026 - Present
Visiting Researcher Innovative Biomaterials Lab	
Visiting appointment scheduled through January 2027	
BLISS Lab Yonsei University	2024 - Present
Graduate Researcher, Integrated M.S./Ph.D. Program	
BLISS Lab Yonsei University	2023 - 2024
Undergraduate Research Assistant	
NCS Lab Yonsei University	2021 - 2023
Undergraduate Research Assistant	

TECHNICAL SKILLS

Programming & development: Python, C++, MATLAB; web and mobile app development

Electronics & embedded systems: STM32, ESP32, Arduino (ATtiny); PCB design, EasyEDA, KiCad

Design & visualization: AutoCAD, Fusion 360, KeyShot, GraphPad Prism, Origin, Adobe Illustrator, ImageJ

Fabrication & instrumentation: SEM, AFM, FTIR, XPS, spin coating, 3D printing, laser cutting

Laboratory: Mammalian cell culture; in vivo studies (murine)

HONORS & LANGUAGES

Best Poster Award | KSBM Spring Meeting, March 2026

Languages: Korean, English

KEUN-YOUNG YOOK

PUBLICATIONS

Journal Articles

K.-Y. Yook et al.

Synergistic Mechano-Fluidic Self-Fusion of Poly(vinyl alcohol)-Modified Liquid-Metal Particles for Post-Treatment-Free Soft Electronics.

Advanced Science, e78062 (2026). Published online 6 October 2026. **First author**. 10.1002/advs.78062

T. Y. Kim, W.-J. Lee, Y. Lee, S. J. Kim, S. Min, S. Chung, S. A. Kim, **K.-Y. Yook**, C.-H. Moon, Y. Lee, K. Park, D.-H. Kim, and J. Seo.

Slippery dopamine-fluoropolymer hybrid surface for improving biliary stent longevity.

Bioactive Materials 61, 210-228 (2026). DOI

D. Lee, S. A. Kim, B.-J. Shim, Y. Lee, T. Y. Kim, S. Park, Y. Lee, H. G. Choi, K. Son, S. B. Han, **K.-Y. Yook**, S. J. Kim, W.-Y. Lee, J. Seo, and J. Kim.

Ion-Mediated Structural Engineering of Hydrogel Interfaces for Tunable Mechanical and Analyte Diffusion Properties in Electrochemical Biosensors.

Advanced Materials 38(19), e15767 (2026). DOI

T. Y. Kim, K. Son, C.-H. Moon, **K.-Y. Yook**, S. A. Kim, H. Ham, Y. Lee, S. I. Lee, Y. Jo, Y. Yu, D.-H. Kim, and J. Seo. Quinone-mediated, tissue-adaptive double-network hydrogel for instant hemostasis and wet-tissue adhesion.

Nature Communications 17, 5526 (2026). DOI

T. Y. Kim, Y. Son, **K.-Y. Yook**, D. G. Lee, Y. Kim, S. J. Kim, K. Park, Y. Lee, T. K. Lee, J. J. Chung, C. Yang, S. Park, and J. Seo.

Bioadaptive liquid-infused multifunctional fibers for long-term neural recording via BDNF stabilization and enhanced neural interaction.

Science Advances 11(37), eadz1228 (2025). DOI

MANUSCRIPTS

K.-Y. Yook et al. "A Simple and Additive-Free Method for Preparing Liquid Metal Paste via Sonication for Robust Interconnects in Soft Electronic Applications." *In preparation; first author*.

CONFERENCE PRESENTATIONS & RECOGNITION

KSBM 2026 - 30th Anniversary International Conference | Busan, Korea | September 2026

"One-Step Solvent-Free Synthesis of an Amorphous-Oxide Percolating Liquid-Metal Paste for System-Level Stretchable Bioelectronics."

K.-Y. Yook: co-first author. Presented by Dr. T. Y. Kim, recipient of the **GSSOC Outstanding Research Presentation Award**.

KSBM Spring Meeting | Seoul, Korea | March 2026

K.-Y. Yook et al. "Synergistic Mechano-Fluidic Self-Fusion of PVA modified Liquid Metal Particles for Sintering-Free Soft Electronics." Poster presentation; **Best Poster Award**.

MRS Fall Meeting & Exhibit | Boston, MA, USA | December 2025

K.-Y. Yook et al. "A Simple and Additive-Free Method for Preparing Liquid Metal Paste via Sonication for Robust Interconnects in Soft Electronic Applications." Oral presentation.

IEIE Summer Annual Conference | Jeju, Korea | 2025

K.-Y. Yook et al. "One-step Fabrication of Highly Adhesive LM Paste: High-Resolution Printing for Soft Electronics." Poster presentation.